Kleanthis Avramidis

3710 McClintock Ave., RTH 318
Los Angeles, CA 90089, United States

avramidi@usc.edu
klean2050.github.io

RESEARCH INTERESTS

Physiological & Biomedical Signal Processing, Efficient & Grounded Representations, Multimodal Representation Learning, Multimodal Interactions, Health and Wellness

EDUCATION

PhD in Computer Science *08/2021 - 12/2026*
University of Southern California (USC), Los Angeles, CA
Thesis: *Towards Efficient and Grounded Biosignal Representations*
Advisor: Prof. Shrikanth Narayanan

Joint BSc & MEng in Electrical Engineering *10/2015 - 07/2021*
National Technical University of Athens (NTUA), Greece
Thesis: *Affective Analysis and Interpretation of Brain Responses to Music Stimuli*
Advisor: Prof. Petros Maragos

SELECTED PROJECTS

SENTINEL: Neurobehavioral Trajectories in Everyday Life *08/2026 - Present*
USC – UCLA

- Longitudinal characterization of physiological and behavioral trajectories associated with depression, suicidal ideation, and resilience.
- Integration of repeated laboratory assessments with wearable and smartphone sensing to examine changes across contexts and time.

Entrainment in Expert and Novice Musicians *07/2025 - Present*
USC Center for Music, Brain, and Society

- Methods to quantify and compare neurophysiological entrainment to music
- Identification of differences in attentional strategies and emotional responses
- Predominant musical elements in driving entrainment intensity

Speech Paralinguistics from Facial Muscle Activity *09/2024 - Present*
USC – Technical University of Munich (TUM)

- Developing machine learning models for translating between EMG activations and audio speech waveforms with application to silent speech interfaces
- Analysis of the temporal dynamics of muscle activation in speech production
- Affective elements of speech in vocal versus silent articulation

PRECOG: Multimodal Integration of Neural and Biobehavioral Signals for Predicting Preconscious Responses *05/2023 - 08/2025*
USC – UCLA

- Developing representation learning models for brain activity signals (EEG)
- Analysis of human physiology (ECG, GSR, eye-tracking) in controlled settings
- Inference on detecting biomarkers of depression and suicidal ideation
- Applied trial-based bootstrapping to regularize deep learning modeling of short experimental trials, applied on eye tracking; first-authored 2 articles

Sensor Fusion for Stress Detection in Driving *05/2022 - 06/2025*
USC – Toyota Research Institute NA – MIRISE

- Developing methods for sensor fusion & self-supervision on physiological signals
- Coordinating multiple data collection processes in the driving setting
- Applied methods for Time-Series Segmentation and Clustering to detect change points in drivers' affective state; authored 2 articles

CVI Assessment through Eye Tracking *02/2022 - 06/2024*
USC – Children's Hospital Los Angeles

- Designing maps of visual saliency on stimulus images to assess differences of Cortical Visual Impairment (CVI) cases against controls

Automatic Differentiation of Pediatric Papilledema *02/2022 - 08/2023*
USC – Children's Hospital Los Angeles – External Sites

- Building deep learning models to differentiate Papilledema from pseudo-cases in challenging pediatric cases, with data collected from multiple sites
- Contributed 1 publication and 2 abstracts within an interdisciplinary team

Wearable Sensing for Family Well-being *10/2021 - 08/2023*
UT Austin – Texas A&M – USC

- Configured scripts to clean and process raw data from multiple wearable sensors
- Leading the development of statistical and learning methods to identify predictive elements of family reported well-being and conflict
- Contributing and cooperating with collaborators from the Psychology field
- Expanding methodologies to relevant projects on assessing workplace stress

WORK EXPERIENCE

Signal Analysis and Interpretation Lab *08/2021 - Present*
University of Southern California, Los Angeles, CA
Graduate Research Assistant, under Prof. Shrikanth Narayanan

- Building a multi-step training framework for audiovisual learning of music representations from official video clips, authored 1 abstract and poster
- Coordinator of project-wise lab and colab meetings, research mentor of undergraduate and master's students from the US, Denmark, and Greece

Audio & Acoustics Research Group *05/2026 - 08/2026*
Microsoft Corporation, Redmond, WA
Research Intern, under Dimitra Emmanouilidou, PhD

- Developed foundation models for advancing brain-computer interfaces
- Proposed evaluation methods to better interpret models trained on biosignals

Chair of Health Informatics *05/2025 - 06/2025*
Technical University of Munich, Germany
Visiting Doctoral Researcher, under Prof. Björn Schuller

- Contributed to a study on Silent Paralinguistics, including data collection design and analysis of surface electromyography (EMG) during speech production

Meta Reality Labs *05/2024 - 08/2024*
Meta Platforms Inc., New York City, NY
Research Scientist Intern, under Nariman Farsad, PhD

- Developed AI methods for advancing augmented reality experiences
- Created machine learning models for decoding neuromotor signals

Toyota Research Institute of North America *05/2023 - 08/2023*
Toyota Motor North America, Ann Arbor, MI
Research and Development Co-op, under Paul Schmalenberg, MSc

- Developed methods for sensor fusion & anomaly detection on biosignals
- Created machine learning models for contact-less heart rate estimation
- Coordinated machine learning software for physics-informed AI applications

Computer Vision, Speech & Signal Processing Lab *07/2019 - 07/2021*
National Technical University of Athens, Greece
Undergraduate Research Assistant, under Prof. Petros Maragos

- Conducted Research in Musical Instrument Recognition
  Co-authored 2 publications, completed my MEng Diploma Thesis

SKILLS

Tools and Libraries: Python, LaTeX, Git, Jupyter, PyTorch, Pandas, Scipy, Librosa, scikit-learn, scikit-image, OpenCV, Transformers, Neurokit2, MNE-Python

Service: IEEE (Graduate Student Membership, Reviewer: ICASSP, ISBI, IJCNN, WASPAA, J-BHI, TBME), ACM (Student Membership, Reviewer: TOMM, MM), Nature (Reviewer: Biomedical Engineering, npj Digital Medicine, Scientific Reports), ISCA (Reviewer: Interspeech, SLT), Elsevier (Reviewer: BSP-C)

Languages: Greek (native), English (proficient), German (intermediate)

MENTORSHIP

- **Andreas Peter Juhl Hansen**, MSc at Aalborg University *2025 – 2026*
- **Simon Bock Segaard**, MSc at Aalborg University *2025 – 2026*
- **Jeppe Roden Münster**, MSc at Aalborg University *2025 – 2026*
  then PhD at Aalborg University
- **Eileen Yang**, MSc at USC *2025 – 2026*
- **Ioanna Chourdaki**, MEng at NTUA *2024 – 2025*
  then PhD Student at USC
- **Angelly Cabrera**, BSc at USC *2023 – 2024*
  then Associate Researcher at NIST
- **Kranti Adsul**, MSc at USC *2023*
- **Shanti Stewart**, MSc at USC *2022*
  then PhD Student at Queen Mary University of London

HONORS AWARDS

☆ **USC Center for Music, Brain, and Society Pilot Grant** *06/2025*
to investigate neural entrainment to music in expert and novice musicians

☆ **USC ECE Research Festival – Outstanding Poster Award** *11/2024*

☆ **USC GSG Professional Development Fund** *05/2024*
Travel award towards attending IEEE EMBC 2024

☆ **IEEE EMBC NextGen Scholar Award** *04/2024*
Grant towards conference registration at IEEE EMBC 2024

☆ **Gerondelis Foundation Scholarship** *06/2023*

☆ **Best Poster Award**: Artificial Intelligence to Classify Fundus Photographs of Pediatric Pseudopapilledema and Papilledema, 48th AAPOS Annual Meeting *05/2023*

☆ **Future Vision Forum Award** *10/2022*
Acceptance and grant to participate with a poster presentation at
invitation-only Symposium of Human-Centric Computing in Ophthalmology

☆ **Oxford Summer School in Machine Learning 2022** *08/2022*
Accepted to participate at the Machine Learning for Healthcare track

☆ **Second Prize @ NEUROHACK 2022** *01/2022*
Award for a Machine Learning model predicting Dementia

☆ **Computer Science PhD Fellowship** *08/2021*
University of Southern California

☆ **A Great Moment for Education** *01/2016*
Award and Grant from Eurobank EFG for the highest
University Entrance Examination Score of my High School

TALKS EVENTS

1. **CuttingEEG Local Garden Symposium**
   Talk: "Scaling Psychiatric Biomarkers in Context and Time"
   September 2026, San Diego, CA
2. **Apple Workshop on Machine Learning for Health**
   Poster: "Scaling Representation Learning from Ubiquitous ECG with SSMs"
   October 2025, Cupertino, CA
3. **Chair of Health Informatics, Technical University of Munich**
   Talk: "Biobehavioral Signal Processing for Mental Health Understanding"
   June 2025, Munich, Germany
4. **Mapping Brain-Body-Behavior Signal Dynamics**
   Event: Satellite Workshop at ICASSP 2025 (Student Volunteer)
   April 2025, Hyderabad, India
5. **Society for Affective Science Annual Conference (SAS 2023)**
   Talk: "Psychophysiology Sensing via Wearables to model Family Well-being"
   March 2023, Long Beach, CA
6. **2022 Future Vision Forum: Human-Centric Computing**
   Poster: "Deep Learning Modeling for Papilledema Diagnosis"
   November 2022, Los Angeles, CA
7. **International Society for Research on Emotion (ISRE 2022)**
   Poster: "Context-aware Representations of Affect in Media from Music and Visual Streams: A Self-supervised Approach"
   July 2022, Los Angeles, CA

TEACHING

1. **CSCI 698: Practicum in Teaching Computer Science** *Fall 2026*
   University of Southern California (Teaching Assistant)
2. **CSCI 467: Foundations of Machine Learning** *Spring 2026*
   University of Southern California (Teaching Assistant)

JOURNAL PUBLICATIONS

J1. **K. Avramidis**, W. Jeong, A. Kommineni, and others, and S. Narayanan
"Deep Learning Characterizes Depression and Suicidal Ideation in Young Adults from Eye Movements".
npj Digital Medicine, 2026.

J2. W. Jeong, A. Kommineni, **K. Avramidis**, and others, and R. M. Leahy
"Time-resolved EEG Decoding Reveals Altered Neural Dynamics of Affective Semantic Evaluation in Depression and Suicidality".
Communications Biology, 2026.

J3. A. Kommineni, W. Jeong, **K. Avramidis**, and others, and S. Narayanan
"Neural Responses to Affective Sentences Reveal Signatures of Depression".
Translational Psychiatry, 2026.

J4. S. R. Kadiri, B. R. Cahn, T. Medani, C. McDaniel, A. Kommineni, **K. Avramidis**, and others, and S. Narayanan
"Implicit Death Associations in Young Adults: Evaluating the Death-Brief IAT for Differentiating Mental Health Status".
Frontiers in Psychiatry, vol. 17, 2026.

J5. C. McDaniel, M. L. Hughes, and others, **K. Avramidis**, and others, and A. Habibi
"Neural Evidence of Disrupted Self-Referential Processing in Suicidal Depression".
Journal of Affective Disorders, 2026.

J6. D. Kunc, J. Komoszyńska, **K. Avramidis**, T. Feng, S. Narayanan, P. Kazienko, and S. Saganowski
"Dominance and Complementarity in Cross-Modal Representation Learning for Wearable Time Series".
Information Fusion, 2026.

J7. P. D. Schmalenberg, H. Hamidi Shishavan, P. Singh, **K. Avramidis**, F. M. Q. Severgnini,
B. Pardo, T. Lee, and E. M. Dede
"Non-invasive Cardiac Sensing via an Acoustic Helmholtz Resonator Cavity with Electrocardiogram Waveform Reconstruction".
Communications Engineering, 2026.

J8. K. Matsunaga, **K. Avramidis**, M. S. Borchert, S. Narayanan, and M. Y. Chang
"Generative Artificial Intelligence (AI)-enabled Saliency Analysis of Eye Tracking in Cerebral/Cortical Visual Impairment (SET-CVI)".
Ophthalmology Science, 2026.

J9. A. C. Timmons, A. A. Tutul, **K. Avramidis**, and others, and T. Chaspari
"Developing Personalized Algorithms for Sensing Mental Health Symptoms in Daily Life".
npj Mental Health Research, 2025.

J10. K. Matsunaga, **K. Avramidis**, M. S. Borchert, S. Narayanan, and M. Y. Chang
"Method for Assessing Visual Saliency in Children with Cerebral/Cortical Visual Impairment using Generative Artificial Intelligence".
Frontiers in Human Neuroscience, 2025.

J11. M. Y. Chang, G. Heidary, and others, **K. Avramidis**, M. Rostami, and S. Narayanan
"Artificial Intelligence to Differentiate Pediatric Pseudopapilledema and True Papilledema on Fundus Photographs".
Ophthalmology Science, 2024.

J12. **K. Avramidis**, D. Kunc, B. Perz, and others, and S. Narayanan
"Scaling Representation Learning from Ubiquitous ECG with State-Space Models".
IEEE Journal of Biomedical and Health Informatics (J-BHI), 2024.

CONFERENCE PUBLICATIONS

C1. M. Ma, E. Zhou, L. Ludwig, and others, **K. Avramidis**, and others, and S. Narayanan
"Smiling Regulates Emotion During Traumatic Recollection".
Proc. ACM Int'l Conf. on Multimodal Interaction (ICMI), 2026.

C2. N. Mehlman, K. Zareno, **K. Avramidis**, A. Xu, and S. Narayanan
"Speech Entrainment in Multi-Party Conversations with a Digital Agent".
Proc. Interspeech, 2026.

C3. A. Kommineni, E. Zhou, **K. Avramidis**, T. Feng, and S. Narayanan
"A Multi-dimensional Framework for Evaluating Generalization in EEG Foundation Models".
Proc. Conf. on Health, Inference, and Learning (CHIL), 2026.

C4. **K. Avramidis**, M. Hughes, and others, and S. Narayanan
"A Point Process Model of Skin Conductance Responses in a Stroop Task for Predicting Depression and Suicidal Ideation".
Proc. Int'l Conf. on Acoustics, Speech and Signal Processing (ICASSP), 2026.

C5. M. Ma, J. Prescott, E. Zhou, T. Feng, **K. Avramidis**, G. M. Tóth, and S. Narayanan
"Encoding Emotion Through Self-Supervised Eye Movement Reconstruction".
Proc. Int'l Conf. on Acoustics, Speech and Signal Processing (ICASSP), 2026.

C6. J. Lee, K. Huang, **K. Avramidis**, and others, and S. Narayanan
"Articulatory Feature Prediction from Surface EMG during Speech Production".
Proc. Interspeech, 2025.

C7. A. Kommineni, **K. Avramidis**, R. Leahy, and S. Narayanan
"Knowledge-guided EEG Representation Learning".
Proc. IEEE Engineering in Medicine and Biology Conference (EMBC), 2024.

C8. **K. Avramidis**, M. Y. Chang, R. Sharma, M. S. Borchert, and S. Narayanan
"Evaluating Atypical Gaze Patterns through Vision Models: The Case of Cortical Visual Impairment".
Proc. IEEE Engineering in Medicine and Biology Conference (EMBC), 2024.

C9. J. Lee, A. Kommineni, T. Feng, **K. Avramidis**, X. Shi, S. Kadiri, and S. Narayanan
"Toward Fully-End-to-End Listened Speech Decoding from EEG Signals".
Proc. Interspeech, 2024.

C10. S. Stewart, **K. Avramidis**, T. Feng, and S. Narayanan
"Emotion-Aligned Contrastive Learning Between Images and Music".
Proc. Int'l Conf. on Acoustics, Speech and Signal Processing (ICASSP), 2024.

C11. **K. Avramidis**, S. Stewart, and S. Narayanan
"On the Role of Visual Context in Enriching Music Representations".
Proc. Int'l Conf. on Acoustics, Speech and Signal Processing (ICASSP), 2023.

C12. **K. Avramidis**, K. Adsul, D. Bose, and S. Narayanan
"Signal Processing Grand Challenge 2023 – e-Prevention: Sleep Behavior as an Indicator of Relapses in Psychotic Patients".
Proc. Int'l Conf. on Acoustics, Speech and Signal Processing (ICASSP), 2023.

C13. **K. Avramidis**, T. Feng, D. Bose, and S. Narayanan
"Multimodal Estimation of Change Points of Physiological Arousal During Driving".
Proc. Int'l Conf. on Acoustics, Speech and Signal Processing Workshops (ICASSPW), 2023.

C14. **K. Avramidis**, M. Rostami, M. Chang, and S. Narayanan
"Automating Detection of Papilledema in Pediatric Fundus Images with Explainable Machine Learning".
Proc. Int'l Conf. on Image Processing (ICIP), 2022.

C15. **K. Avramidis**, C. Garoufis, A. Zlatintsi, and P. Maragos
"Enhancing Affective Representations of Music-Induced EEG through Multimodal Supervision and Latent Domain Adaptation".
Proc. Int'l Conf. on Acoustics, Speech and Signal Processing (ICASSP), 2022.

C16. **K. Avramidis**, A. Zlatintsi, C. Garoufis, and P. Maragos
"Multiscale Fractal Analysis on EEG Signals for Music-Induced Emotion Recognition".
Proc. European Signal Processing Conference (EUSIPCO), 2021.

C17. **K. Avramidis**, A. Kratimenos, C. Garoufis, A. Zlatintsi, and P. Maragos
"Deep Convolutional and Recurrent Networks for Polyphonic Instrument Classification from Monophonic Raw Audio Waveforms".
Proc. Int'l Conf. on Acoustics, Speech and Signal Processing (ICASSP), 2021.

C18. A. Kratimenos, **K. Avramidis**, C. Garoufis, A. Zlatintsi, and P. Maragos
"Augmentation Methods on Monophonic Audio for Instrument Classification in Polyphonic Music".
Proc. European Signal Processing Conference (EUSIPCO), 2020.

PREPRINTS

P1. **K. Avramidis**, I. Tashev, and D. Emmanouilidou
"BrainCodes: A Lightweight EEG Foundation Model over Discrete Neural Codes".
arXiv preprint arXiv:2610.29320, 2026.

P2. **K. Avramidis**, I. Tashev, and D. Emmanouilidou
"Probing Spatial and Spectral Grounding in EEG Foundation Models".
arXiv preprint arXiv:2610.29367, 2026.

P3. T. Chae, **K. Avramidis**, D. Emmanouilidou, and I. Tashev
"MtReNIRS: A Self-Supervised Foundation Model for Transferable Neurovascular Representations".
arXiv preprint arXiv:2610.36159, 2026.

P4. J. Prescott, **K. Avramidis**, and S. Narayanan
"Adapting Appearance-Based Gaze Estimation to Narrow-Range, Long-Duration Screen Viewing".
arXiv preprint arXiv:2610.11036, 2026.

P5. J. Lee, **K. Avramidis**, J. Lee, T. Feng, N. Dehak, and S. Narayanan
"MyoCodec: A Streaming Neural Codec for Electromyography".
arXiv preprint arXiv:2609.36687, 2026.

P6. A. Kommineni, E. Zhou, **K. Avramidis**, and others, and S. Narayanan
"Aperiodic and Low-Frequency Spectral Bias in Reconstruction-based EEG Foundation Models".
arXiv preprint arXiv:2605.26434, 2026.

P7. E. Zhou, M. Ma, **K. Avramidis**, G. M. Tóth, and S. Narayanan
"Looking Into the Past: Eye Movements Characterize Elements of Autobiographical
Recall in Interviews with Holocaust Survivors".
arXiv preprint arXiv:2604.22016, 2026.

P8. S. Pistrosch, **K. Avramidis**, Z. Ren, and others, and B. W. Schuller
"Affect Decoding in Phonated and Silent Speech Production from Surface EMG".
arXiv preprint arXiv:2603.11715, 2026.

P9. T. Feng, **K. Avramidis**, A. Xu, D. Wang, B. M. Booth, and S. Narayanan
"VoxCare: Studying Natural Communication Behaviors of Hospital Caregivers through Wearable Sensing of Egocentric Audio".
arXiv preprint arXiv:2603.10888, 2026.

P10. I. Chourdaki, **K. Avramidis**, C. Garoufis, A. Zlatintsi, and P. Maragos
"Teager-Kaiser Energy Methods for EEG Feature Extraction in Biomedical Applications".
arXiv preprint arXiv:2511.17164, 2025.

P11. W. Jeong, W. Cui, **K. Avramidis**, T. Medani, S. Narayanan, and R. Leahy
"Informed Bootstrap Augmentation Improves EEG Decoding".
arXiv preprint arXiv:2511.12073, 2025.

P12. **K. Avramidis**, T. Feng, W. Jeong, J. Lee, W. Cui, R. M. Leahy, and S. Narayanan
"Neural Codecs as Biosignal Tokenizers".
arXiv preprint arXiv:2510.09095, 2025.

P13. **K. Avramidis**, E. Zhou, T. Feng, and others, and S. Narayanan
"Tracking Driving Stressors through Multimodal Physiological Monitoring".
arXiv preprint arXiv:2507.14146, 2025.

P14. M. Hughes, C. McDaniel, and others, **K. Avramidis**, and others, and A. Habibi
"PRECOG: Comprehensive Experimental Design and Stimuli Protocol for Investigating Depression and Suicidal Ideation".
Open Science Framework (OSF) preprint, 2025.

P15. A. Cabrera, **K. Avramidis**, and S. Narayanan
"Early Detection of Coffee Leaf Rust through Convolutional Neural Networks Trained on Low-Resolution Images".
arXiv preprint arXiv:2407.14737, 2024.

ABSTRACTS

A1. M. Chang, M. W. Reid, K. Matsunaga, **K. Avramidis**, and S. Narayanan
"Eye Tracking Metrics Mediate Adaptive Behavior Challenges in Children with Autism and Co-morbid Visual Disorders".
Investigative Ophthalmology & Visual Science (ARVO), vol. 67, no. 7, 2026.

A2. K. Matsunaga, **K. Avramidis**, S. Narayanan, M. Borchert, and M. Chang
"Assessing Visual Saliency in Children with Cerebral/Cortical Visual Impairment (CVI)
Using Generative AI and Eye Tracking".
Journal of the American Association for Pediatric Ophthalmology and Strabismus (AAPOS), 2025.

A3. M. Chang, **K. Avramidis**, R. Sharma, M. Borchert, and S. Narayanan
"Saliency Analysis of Eye Tracking in Children with Cortical/Cerebral Visual Impairment (CVI) Enabled by Machine Learning".
Investigative Ophthalmology & Visual Science (ARVO), vol. 65, no. 7, 2024.

PATENTS

T1. F. M. Q. Severgnini, **K. Avramidis**, P. Singh, E. M. Dede, and P. D. Schmalenberg
"Systems and Methods for Vehicle Biometrics Processing and Vehicle Control".
U.S. Patent Application 18/817,888 (Pub. No. US 2026/0062019 A1), 2026.

T2. P. D. Schmalenberg, **K. Avramidis**, F. M. Q. Severgnini, and B. Pardo
"Systems and Methods for Noncontact Monitor of Cardiac Activities".
U.S. Patent Application 18/374,964 (Pub. No. US 2025/0107770 A1), 2025.